QUIZ – QUESTION PAPER

Course Code	EEE418		
Course Name	Renewable Energy		
Lecturer	Dr. Abdul Khaliq Rasheed		
Academic Session	2025/09		
Duration	1 hour	Student ID	

- **Answer all the following five questions. (Total = 10 marks)**
- **Each answer gets 2 marks.**
- **Write clearly and in detail.**

1. **(2 Marks)**

A wind farm developer is considering two sites: a remote onshore mountain pass and a shallow offshore location. **Compare** the primary **environmental impact** and the major **Capital Expenditure (CAPEX) driver** for developing these two distinct sites.

For the environmental impact, contrast the type of disruption (e.g., land vs. marine ecosystems). For the CAPEX driver, consider the primary component that accounts for the highest cost difference between the two site types (e.g., foundations, grid connection, or turbine installation).

2. **(2 Marks)**

Analyze the operational significance of the **cut-in speed** and the **cut-out speed** for a modern utility-scale wind turbine. **Explain** why the turbine must shut down at the cut-out speed.

Clearly define the purpose of both speeds in relation to energy production and safety. When explaining the shutdown, relate it to the excessive aerodynamic forces and the need to protect the turbine's structural integrity (e.g., blades and gearbox).

3. **(2 Marks)**

Describe the key difference between **Flash Steam** and **Binary Cycle** geothermal power plants in terms of the **working fluid** used in the power generation loop and the **minimum temperature requirement** of the geothermal resource.

Note: State whether the geothermal fluid is used directly or if a secondary fluid is involved. Link this choice of technology directly to the required temperature of the resource to ensure efficient operation.

4. **(2 Marks)**

Analyze the primary energy efficiency challenge of **green hydrogen** production based on the entire Power-to-Gas-to-Power chain (electricity to hydrogen back to electricity). **Explain** why this overall efficiency loss is often considered an acceptable trade-off for **seasonal renewable energy storage**.

Note: Identify the typical efficiency range for the full Power-to-Power cycle and the stages where the largest losses occur. Justify the low efficiency by discussing the value proposition of hydrogen (e.g., high energy density, long duration storage) compared to alternatives.

5. **(2 Marks)**

Differentiate Pumped Hydro Storage (PHS) from **Battery Energy Storage Systems (BESS)** based on their typical **storage duration/capacity** and their respective **siting constraints**.

Note: Use technical terms (like MWh or GWh) to quantify the difference in capacity. For siting, explain the topographical and geographical requirements of PHS versus the modularity and installation flexibility of BESS.

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